

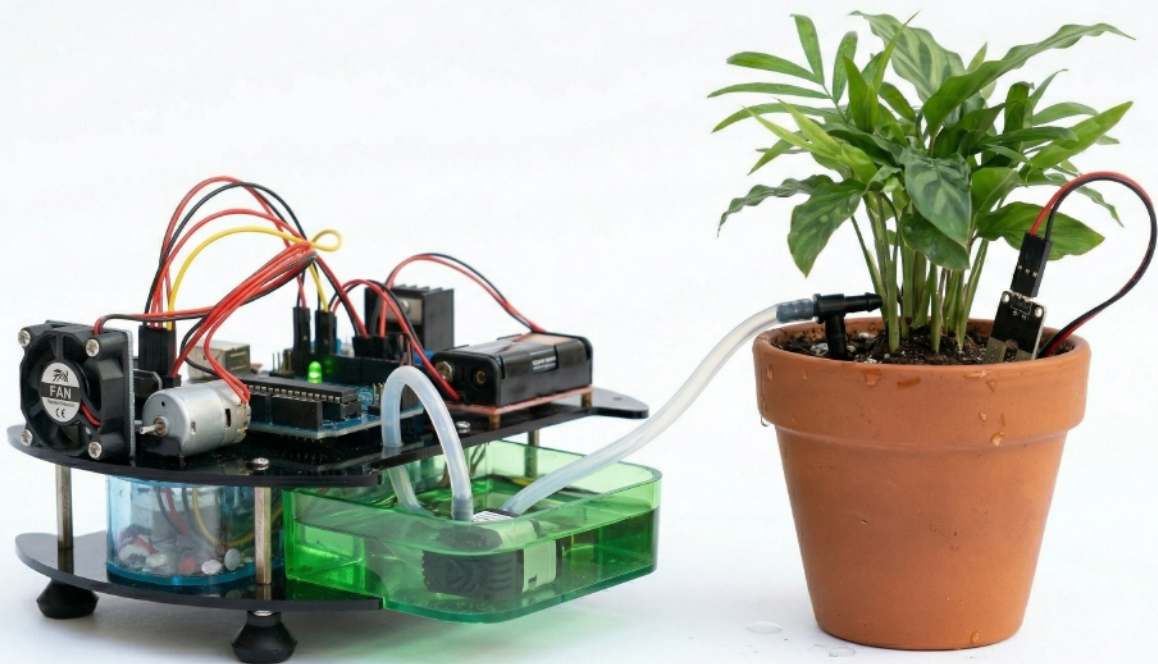


Society of Agricultural
Robotics Engineers



CTRL LABS BUILD GUIDE

AUTOMATIC PLANT WATERING SYSTEM



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01 OVERVIEW

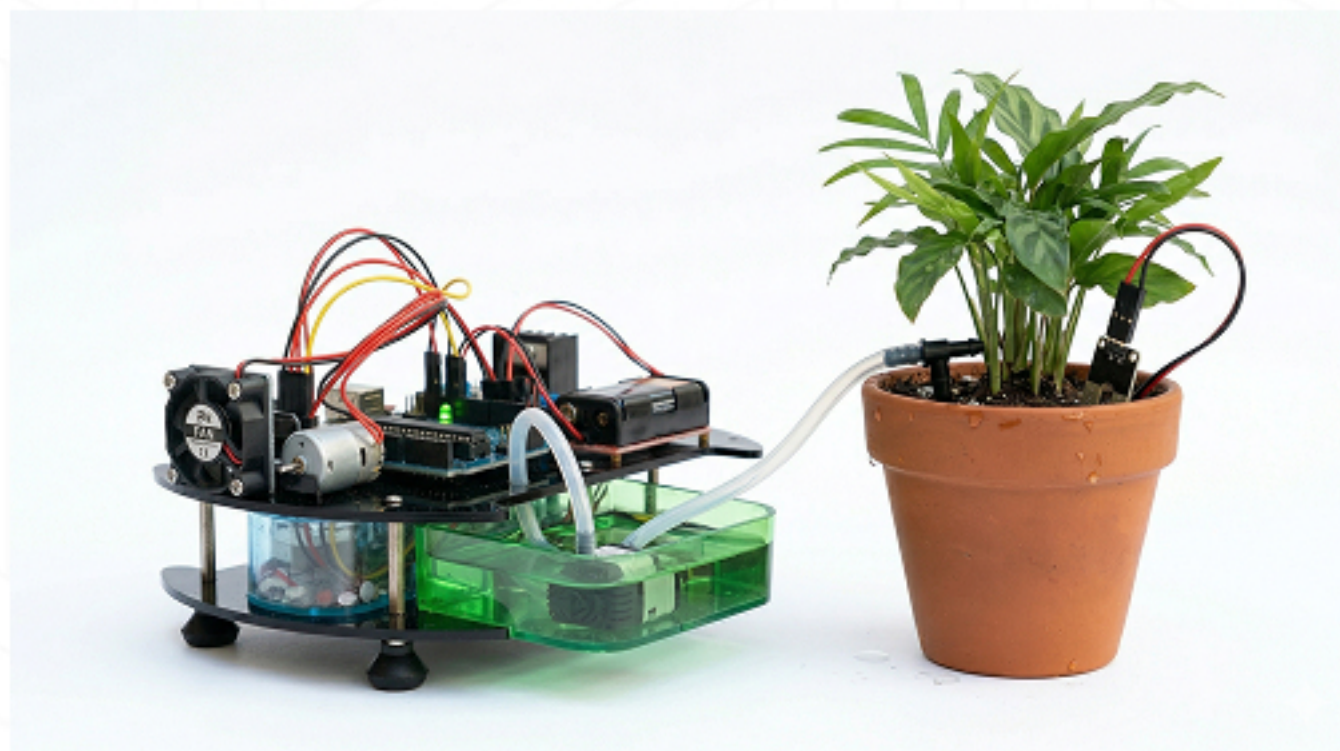
AN AUTOMATIC PLANT WATERING SYSTEM

The IoT Automatic Plant Watering System is a smart irrigation solution designed to monitor soil moisture and automatically water plants whenever the soil becomes too dry. The system uses an ESP8266 as the main controller, allowing real-time monitoring through a web dashboard.

Control & IoT System

Soil Monitoring System

Water level Monitoring System



Objectives:

- Understand the basics of IoT systems
- Learn how sensors interact with microcontrollers

• Overview

Components

Circuit

Procedure

Program

**MICROCONTROLLER****ESP8266**

Controls the system and enables IoT communication

**SENSOR****RESISTIVE SOIL MOISTURE SENSOR**

Responsible for detecting soil moisture conditions.

**SENSOR****WATER LEVEL SENSOR**

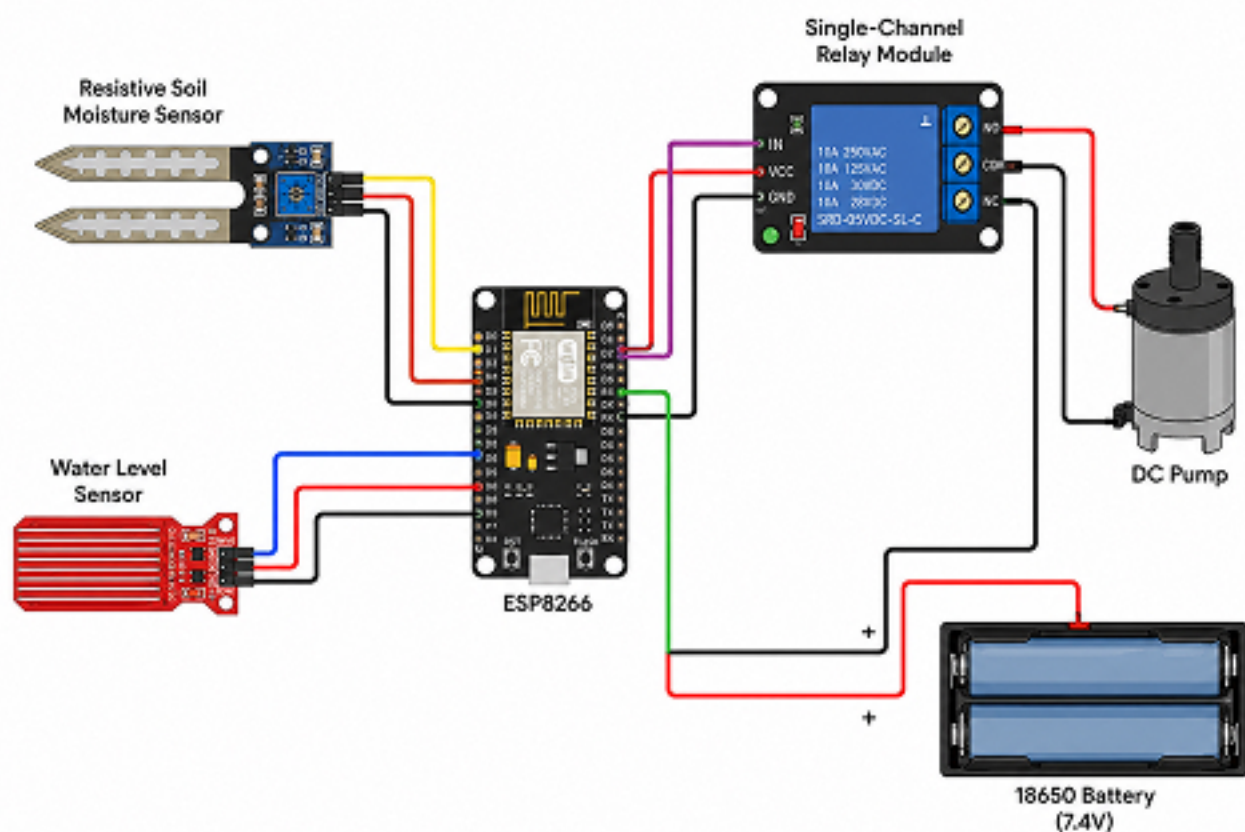
Monitors water level inside the reservoir

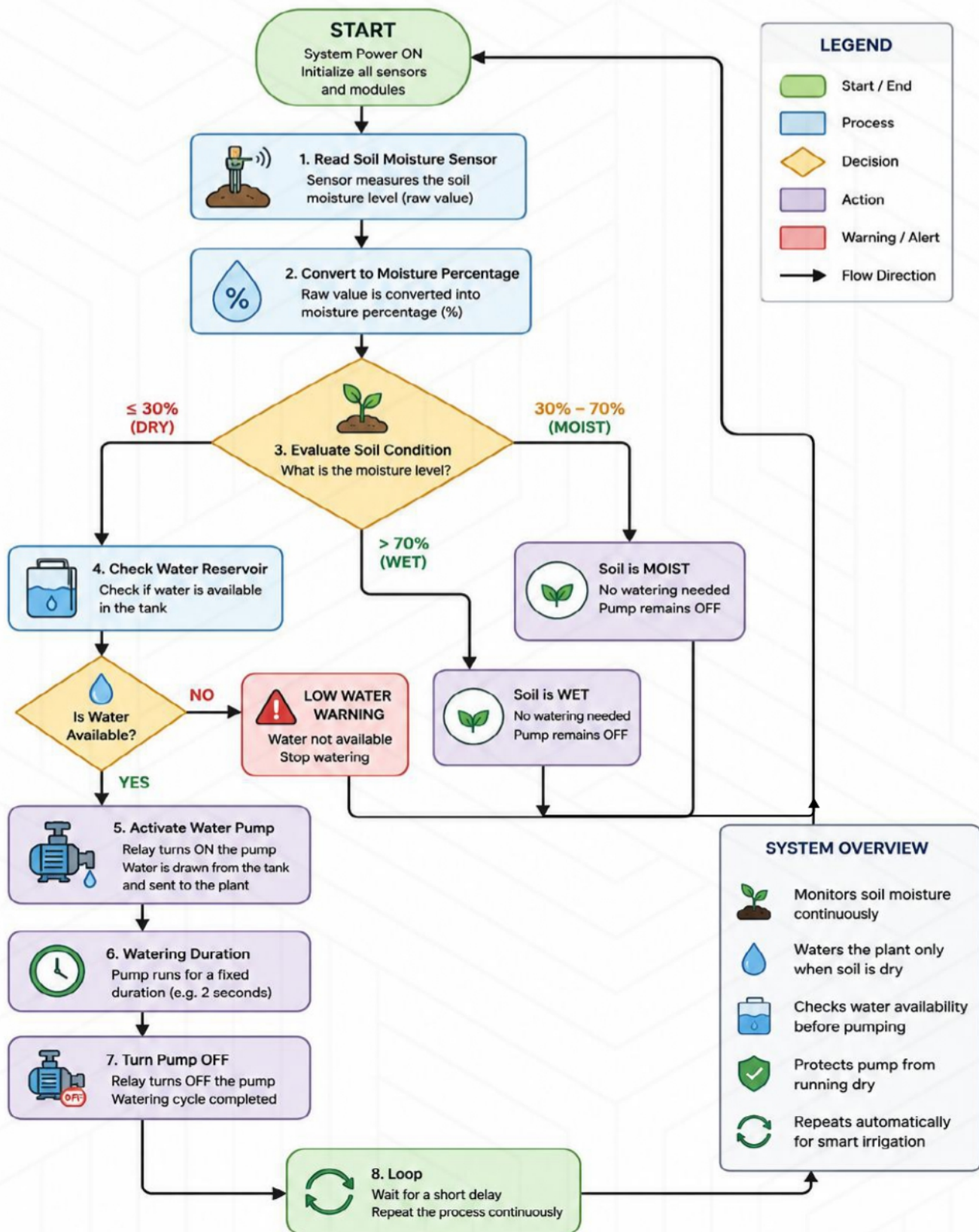
**Relay****SINGLE CHANNEL RELAY**

Safely switches the water pump ON and OFF

**MOTORS & MOTION COMPONENTS****5V DC SUBMERSIBLE PUMP**

Responsible for transporting water from the reservoir to the plant.





04 PROCEDURE

4.1 SENSOR DATA

- Soil Moisture
- Water Level

4.2 WORKING PRINCIPLE

1. The soil moisture sensor continuously checks the moisture level of the soil.
2. If the moisture level falls below the preset threshold, the ESP8266 activates the relay module.
3. The relay module switches ON the water pump.
4. Water is pumped from the reservoir through the tube pipe and dripper to the plant.
5. Once the soil moisture reaches an acceptable level, the ESP8266 turns the pump OFF.
6. The water level sensor monitors the amount of water remaining inside the reservoir.
7. Sensor data is sent to the IoT dashboard for remote monitoring.

4.3 FEATURES

- Automatic plant watering
- Real-time soil moisture monitoring
- Reservoir level monitoring
- IoT dashboard integration
- Manual pump activation
- Smart watering recommendations
- Real-time pump activity monitoring
- Online/offline communication status monitoring

4.4 IOT DASHBOARD FEATURES

- View live soil moisture values
- Monitor moisture status (Dry, Moist, Wet)
- View current pump status
- Monitor watering activity history
- View reservoir level visually
- Monitor communication status of the ESP8266
- Manually activate watering remotely

Overview Live Water Components Schematic Circuit Board on **• Procedure** Pins Program

05 CODE & MODULES

INSTALLABLE LIBRARIES

<NewPing.h>
<Servo.h>

AVAILABLE CODES

- CTRL_LABS-AutoPlantWateringSystem.ino
- DigitalWaterlevelread.ino
- PumpRelayCode.ino
- SoilmoisturesensorReading.ino

 Codes have been attached on the CTRL LABS Platform

4.3 POSSIBLE IMPROVEMENTS

-]Solar-powered system
- Mobile app integration
- Weather-based irrigation
- Multiple plant monitoring
- AI-based watering prediction
- Notification system
- Battery backup system.

4.4 LEARNING OUTCOMES

After completing this project, learners should understand:

- Sensor integration
- Relay control systems
- Basic automation logic
- IoT communication systems
- Real-time monitoring
- Smart agriculture concepts
- Embedded systems basics

Overview

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